12TH CHEMISTRY · CHAPTER 31 · SOLVED EXERCISE

Materials and Energy

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. MULTIPLE CHOICE QUESTIONS

I) A mining engineer discovers a new deposit of a metal that occurs primarily as a carbonate. Which sequence is most efficient for the extraction of metals?

(a) Mining, Concentration, Roasting, Reduction with Carbon
(b) Mining, Concentration, Calcination, Reduction
(c) Mining, Concentration, Electrolysis
(d) Mining, Roasting, Refining

Answer: (b) Mining, Concentration, Calcination, Reduction — A carbonate ore is converted to the oxide by calcination and then reduced.

II) he extraction of aluminium from its oxide is done by electrolysis and not by calcination because:

(a) Carbon forms alloy
(b) Low melting point
(c) Aluminium is more reactive than carbon
(d) Removes gangue

Answer: (c) Aluminium is more reactive than carbon — Carbon cannot reduce the oxides of metals more reactive than itself.

III) Which property makes ceramics suitable for furnace bricks?

(a) Conductivity (b) Low density
(c) High melting point (d) Flexibility

Answer: (c) High melting point — Ceramics have high melting temperatures and resist heat.

IV) In fiberglass, the matrix mainly:

(a) Provides strength (b) Conducts electricity
(c) Holds reinforcement (d) Increases weight

Answer: (c) Holds reinforcement — The matrix holds the reinforcement; the reinforcement provides strength.

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V) Major limitation of mechanical recycling is:

(a) High energy use
(b) Polymer degradation
(c) Monomer formation
(d) use of Na_2CO_3

Answer: (b) Polymer degradation — It is limited by contamination and degradation of polymer chains.

VI) The mobile phone recycling is difficult because of:

(a) Non-recyclable material (b) Mixed materials
(c) No heat sensitivity of materials (d) High melting glass

Answer: (b) Mixed materials — Phones are assembled from mixed materials that are not easily separated.

VII) Chemicals from coal used in dyes are:

(a) Methane series
(b) Benzene, toluene, xylene
(c) Alkenes
(d) CO_2 and H_2

Answer: (b) Benzene, toluene, xylene — Coal gives benzene, toluene and xylene for dyes, resins and explosives.

VIII) Nuclear fission involves:

(a) Combining nuclei (b) Splitting nuclei
(c) Breaking bonds (d) Molecules

Answer: (b) Splitting nuclei — A heavy nucleus is split into two lighter nuclei.

The stem of MCQ II is printed "he extraction..." (the first letter is missing in the textbook) and it says "calcination" where reduction with carbon is meant.

Q2. SHORT ANSWER QUESTIONS

a) What is a chain reaction in nuclear fission?

When a heavy nucleus such as uranium-235 is split by a neutron, it releases more neutrons. These neutrons split further nuclei, which release still more neutrons, so the fission continues by itself. This is called a chain reaction.

b) State the economic challenge associated with reprocessing waste materials.

Reprocessing involves sorting, cleaning and other methods and requires a lot of energy, so it is usually expensive. It becomes uneconomical, especially when the same materials are available elsewhere at a lower price.

c) Keeping in view the concept of microstructure, explain why adding zinc to copper (to form Brass) results in a material that is stronger than pure copper.

In pure copper all the atoms are of the same size and are arranged in regular layers that slide over each other easily. Alloying alters this microstructure: zinc atoms of a different size are distributed among the copper atoms and disturb the regular layers, so the layers cannot slide easily. Brass is therefore stronger (and harder) than pure copper.

d) Define energy density and give one example.

Energy density is the quantity of energy that can be stored per unit volume of a fuel, usually in megajoules per litre (MJ/L). Example: diesel has an energy density of about 35.8 MJ/L (gasoline about 34.2 MJ/L).

e) Name two specific semiconductors mentioned in the text and their modern applications.

(1) Silicon: microchips (e.g. mobile SIM cards) and solar panels. (2) Gallium arsenide (GaAs): wireless communication such as 5G internet devices. (Also gallium nitride, GaN: fast charging in electric cars.)

f) State two applications of X-ray crystallography in science or industry.

(1) Determining the structures of biomolecules, e.g. the double helix of DNA, enzymes and proteins. (2) Pharmaceutical drug development, to see how a drug molecule attaches to its target. (Also: analysis of minerals, synthetic crystals and nanomaterials.)

g) State two advantages and one disadvantage of nuclear fission.

Advantages: (1) very high energy production from a small quantity of fuel; (2) it is used extensively in nuclear power plants to produce electricity (without burning fossil fuels). **Disadvantage:** it produces radioactive waste that must be handled carefully for a long time (there is also the danger of nuclear accidents).

h) In the reduction of Hematite (Fe_2O_3), carbon is added, but carbon monoxide (CO) is the actual species that displaces the oxygen. Explain why the gas (CO) is a more efficient reducing agent in a large-scale furnace than solid carbon (C).

$C + ½O_2 \rightarrow CO$

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Solid carbon can touch the solid iron oxide only at the points where the lumps meet, so the reaction between two solids is slow and incomplete. Carbon monoxide is a gas: it spreads through the whole furnace, passes between and into the lumps of ore and comes into contact with all of the oxide surface. So the reduction is faster and more complete.

Q3. CONSTRUCTED RESPONSE QUESTIONS

a) Explain environmental problem, caused by fossil fuels.

Fossil fuels are highly efficient sources of energy, but their combustion produces greenhouse gases such as carbon dioxide (CO_2) and others. This leads to air pollution, global warming and climate change. They are also non-renewable resources.

Global warming raises the earth's temperature and changes the climate, so the carbon footprint must be reduced by moving to sustainable sources such as solar, hydro and wind energy.

b) Based on the chemical equations provided, explain why roasting sulphide ores poses a greater environmental challenge compared to the calcination of carbonate ores.

$$2ZnS_{(s)} + 3O_{2(g)} \rightarrow 2ZnO_{(s)} + 2SO_{2(g)} \text{ (roasting)}$$
$$ZnCO_{3(s)} \rightarrow ZnO_{(s)} + CO_{2(g)} \text{ (calcination)}$$

Roasting releases sulphur dioxide, SO_2, a toxic, choking gas that pollutes the air and forms acid rain. Calcination releases carbon dioxide, CO_2, which is a greenhouse gas but is not toxic and does not form acid rain. Hence roasting is the greater environmental challenge, and its SO_2 must be trapped.

c) If a sample of copper is 98% pure after reduction, why is the final "refining" step still considered crucial for the manufacturing of electrical wiring?

For electrical wiring copper needs to be 99.99% pure. Even small amounts of impurities reduce its electrical conductivity. So the 98% pure copper must still be refined (electrochemically) to remove the remaining impurities.

d) Explain, why chemical recycling (depolymerization) is often superior to mechanical recycling?

Mechanical recycling	Chemical recycling (depolymerization)
The plastic is sorted, cleaned, melted and reformed.	The polymer chains are broken down into the original monomers or simpler chemicals using heat, solvents or catalysts.
Limited by contamination and by degradation of the polymer chains, so the quality falls.	Produces purer reactants, which can be made into new polymer of good quality.

e) Gold is often found in its free state, while sodium is only found in the combined states. Use the concept of the reactivity series to justify why the extraction process for sodium is significantly more energy-intensive than for gold.

Gold is at the bottom of the reactivity series: it is so unreactive that it does not form compounds easily and occurs as the free metal, so it only has to be mined and separated. Sodium is near the top of the series: it is very reactive, exists only in stable compounds and is more reactive than carbon, so carbon cannot reduce its compounds. It must be extracted by the electrolytic method, which needs a large amount of energy to melt the compound and pass an electric current through it.

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DESCRIPTIVE QUESTIONS

Q4. Explain composite materials by describing and differentiating the roles of the matrix and the reinforcement in them.

A composite is a combination of two or more materials that forms a product with better properties. One component is the **matrix**; the others, caged in the matrix, are the **reinforcement agents** (a composite may have more than one). Example: fibreglass has a polymer matrix reinforced with fibres of fabric, glass or carbon-based materials.

Matrix	Reinforcement
The component in which the other components are caged.	The component(s) caged inside the matrix.
Holds the reinforcement together.	Provides high strength to the material.
In fibreglass: a polymer.	In fibreglass: fibres of fabric, glass or carbon-based materials.

Composites have a good strength-to-weight ratio and strong resistance to corrosion, so they suit demanding applications. Boat hulls and car bodies are made of fibreglass (also used to insulate walls and ceilings from heat and cold); carbon fibre is widely used in aircraft and sports gear because of its lightness and durability.

Q5. Explain how X-rays interact with crystals to produce diffraction patterns and how these patterns help determine atomic positions.

X-ray crystallography is a very strong analytical tool used to identify the three-dimensional positions of atoms in a crystalline solid. X-rays have a very small wavelength, so they can find the positions of electrons and atoms in a molecule or an ionic compound.

1. Solid materials consist of layers or strands of atoms. X-rays from the source fall on the crystal.
2. The rays are diffracted after hitting the atoms at specific angles (path length difference = $2d \sin\theta$).
3. The diffracted rays form patterns on a screen, called diffraction patterns.
4. A diffraction pattern spots the location of atoms and electrons and gives the distances between atoms.
5. Using mathematical models, scientists draw a detailed map of atomic positions, bond angles and electron density, and so determine the overall structure of the molecule or crystal (e.g. naphthalene).

Q6. Describe the main steps involved in the extraction of metals from their ores.

Metallurgy is the science of extracting metals from their ores and making them pure for better use. The reactivity and nature of the metal decide the extraction procedure.

i. Mining: acquiring the metal ores that are buried deep in the earth, e.g. iron ores.

ii. Ore concentration: the ore is made free from impurities (gangue) as much as possible.

iii. Conversion of the concentrated ore to metal oxide: by *roasting* (heating a sulphide ore with an excess supply of air) or by *calcination* (heating a carbonate ore in limited or no air).

$$2ZnS_{(s)} + 3O_{2(g)} \rightarrow 2ZnO_{(s)} + 2SO_{2(g)} \text{ (roasting)}$$
$$ZnCO_{3(s)} \rightarrow ZnO_{(s)} + CO_{2(g)} \text{ (calcination)}$$

iv. Reduction of the metal oxide to metal: for less reactive metals (below carbon in the reactivity series) carbon is the reducing agent. Carbon is first converted into CO, which displaces oxygen from the metal oxide.

$$C + \tfrac{1}{2}O_2 \rightarrow CO$$
$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

Metals more reactive than carbon (Na, K, Al) are obtained by the *electrolytic method*: the metal compound is melted and an electric current is passed; the positive metal ions move to the cathode and gain electrons to form the metal. It needs much energy and is also used to get very high purity in less reactive metals (e.g. Cu).

v. Refining: the final purification of the metal by thermal, chemical or electrochemical methods; e.g. copper must be 99.99% pure for electrical wiring.

vi. Alloying: mixing the metal with other metals or a non-metal to improve its properties.

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

Quick Check 31.1

i. Why ceramics are used as insulators on power lines.

Because most ceramics (e.g. porcelain) do not conduct electricity; they are also hard, durable and resistant to heat and corrosion, so they last in the open air.

ii. Why is silicon preferred for use in microchips and solar panels?

Silicon is a semiconductor that is highly effective and abundant; the current through it can be controlled, and it can conduct, store and process data.

iii. What do reinforcement agents provide in composite materials?

They provide high strength to the material, while the matrix holds them together.

Quick Check 31.2

a) Name two types of hazards caused by toxic materials.

Health hazards and environmental hazards (physical hazards are the third type).

b) Which methods will you use for the recycling i) empty metal food cans ii) empty beverage glass bottles iii) shopping bags and toys

i) Thermal and chemical refining (metallurgy): melting and refining to remove impurities. **ii)** Cullet preparation and melting: cleaning, crushing and melting the scrap glass to form new glass. **iii)** Plastics: mechanical recycling (sorting, cleaning, melting and reforming) or chemical recycling (depolymerization into monomers).

c) How do heavy metals enter the food chain?

Heavy metals such as lead, mercury and cadmium from discarded electronics, batteries and fluorescent lights leach into soil and ground water. From there they are taken up by plants and aquatic organisms such as fish, and pass on to the animals and humans that eat them.

d) How does recycling help conserve natural resources?

Recycled materials are used again in place of fresh raw materials, so limited natural resources such as metals (ores), oil and timber are preserved. It also saves energy, e.g. recycling aluminium uses 95% less energy than extraction from the ore.

Quick Check 31.3

a) If fossil fuels are non-renewable resources, why are they still heavily used worldwide?

Because they are highly efficient sources of energy and their fractions provide transport fuels, electricity, heating and the petrochemicals needed by industry. The existing vehicles, power plants and industries are built to run on them.

b) Can XRD be used to find the structure of liquid?

No. X-ray crystallography needs a crystalline solid, in which atoms are arranged in regular layers that diffract X-rays at specific angles to give a clear pattern. In a liquid the particles are not in a fixed, regular arrangement, so no sharp diffraction pattern is formed.

Quick Check 31.4

a) Why is catalytic cracking more commonly used in modern petroleum refineries?

The catalyst (zeolites or silica-alumina) lowers the activation energy, so cracking takes place at a lower temperature (about 500 °C) and pressure. This saves energy and gives better control over the final products; it is more efficient.

b) If more alkenes such as ethene and propene are desired to be produced from higher hydrocarbons, such as diesel or kerosine, which cracking method should be preferred and why?

Thermal cracking, because it normally generates a high percentage of alkenes.

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c) During reforming, hexane is converted into benzene and hydrogen. Explain why hydrogen can be considered a valuable by-product.

Hydrogen is itself a useful substance: it is a clean fuel with a very high specific energy (120 MJ/kg) and it is needed in industry, e.g. for making ammonia for fertilizers. So it can be collected and used or sold instead of being wasted.

Quick Check 31.5

a) Why are batteries included in many solar energy systems?

To store electricity for use at night or on cloudy days; they also help to stabilise the voltage and provide surge current.

b) Besides power generation, where else is nuclear fission used?

In nuclear weapons (atomic bombs), in the reactors of nuclear submarines and ships, and in research reactors that produce radioisotopes for medicine and industry.

This answer is from general knowledge; the chapter mentions only electricity generation.

c) Identify the missing component in the reaction: $^{235}U + n \rightarrow$ _____ + _____ + 3n + energy

$$^{235}U + {}^{1}n \rightarrow {}^{141}Ba + {}^{92}Kr + 3\,{}^{1}n + \text{energy}$$

The missing components are barium-141 and krypton-92.

d) Why do nuclear reactions release much more energy than chemical reactions?

Chemical reactions involve only the electrons (breaking and making of bonds). Nuclear reactions involve changes in the nucleus, where the forces holding the particles are far stronger; a small amount of mass is converted into a very large amount of energy.

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